

Student Guide

The science, the data, the methods and how your report is marked

HOW TO USE THIS GUIDE

This is the **reference** for the project: keep it open while you work. The **Writing Workbook** has the activities and the tasks between tutorials; the **tutorial slides** have the explanations from each session.

\$1 is the project on one page. **\$2–4** give the science, data and methods you need for your Introduction and Methods. **\$5–6** cover the report itself. **\$7** is the marking scheme.

1. The project at a glance

You will use **ERA5** reanalysis data and the **Oceanic Niño Index (ONI)** to investigate how cloud cover is related to the **El Niño–Southern Oscillation (ENSO)**. After exploring with a guided Jupyter notebook, you choose **one focused question** and answer it in a short scientific report.

The report	Individual; at most 2,000 words ; about 3–5 figures ; 100% of the course mark
Your question	One focused question: a cloud type, region, season or timing, not 'ENSO and all clouds everywhere'
The notebook	A toolbox, not a checklist : use only the analyses that answer your question
Use of AI	AI Minimal : "AI may be used only for limited presentation-related support, such as grammar, fluency, spelling, or formatting. Students' ideas, arguments, and conclusions must remain their own."
Draft	Complete first draft by Thu 3 Dec 2026 ; draft submitted Fri 5 Feb 2027 (formative, no marks, no DASS extensions)
Final report	Fri 12 Mar 2027, 12:00 (DASS extensions apply)

Why it matters

ENSO is the largest source of year-to-year climate variability on Earth. It shifts the odds of drought, flood and tropical storms across much of the planet, and an ENSO forecast is one of the few skilful seasonal forecasts we have. The patterns you will find in ERA5 are the ones national weather services watch. ENSO shifts the odds; it doesn't guarantee any single outcome, and that nuance belongs in your report too.

2. The science you need

2.1 ERA5: a reconstruction of the atmosphere

ERA5 is ECMWF's fifth-generation reanalysis. A weather model is repeatedly corrected towards all available observations (data assimilation), giving a complete, gridded, physically consistent record. We use monthly means on a 0.75° grid for **January 2005 – August 2025**.

Cloud in ERA5 is a model product. Cloud cover is calculated by the model's cloud scheme from its humidity, temperature and vertical motion; observations shape it only indirectly. Trust the **sign** of a change more than its exact size, and say so in your Discussion.

- **TCC, LCC, MCC, HCC:** total, low, medium and high cloud cover, each the fraction (0–1, plotted as %) of a grid box covered at that level.
- **TCC ≠ LCC + MCC + HCC.** Layers overlap, so seen from above the total is less than the sum.
- Changes in cloud cover are in **percentage points (pp)**: 40% → 44% is +4 pp (a 10% relative increase).

2.2 ENSO and the ONI

ENSO is coupled ocean–atmosphere variability in the tropical Pacific. In **El Niño** the central and eastern Pacific is anomalously warm; in **La Niña** it is anomalously cool. The **ONI** is the 3-month running-mean sea-surface temperature anomaly in the Niño 3.4 region (5°N–5°S, 170°W–120°W), from the NOAA Climate Prediction Center. ENSO is **phase-locked** to the seasons: events usually peak around November–January.

2.3 Why clouds change: the Walker circulation

Air rises over the warm western Pacific (deep convection, high cloud), flows east aloft, sinks over the cool eastern Pacific, and returns west as the trade winds. The Maritime Continent always stays the **rising branch**: El Niño **weakens** the circulation, La Niña **strengthens** it. During El Niño, convection and high cloud **shift towards the central Pacific** and are **suppressed over the Maritime Continent**. Any extra cloud further east is usually thin. **Low cloud** responds through different physics (stability, subsidence), so don't assume it follows the high cloud.

Why ENSO changes the strength of tropical cloud
The Maritime Continent is always the rising branch and the eastern Pacific the sinking branch. El Niño weakens that circulation (and can add thin cloud further east); La Niña strengthens it.

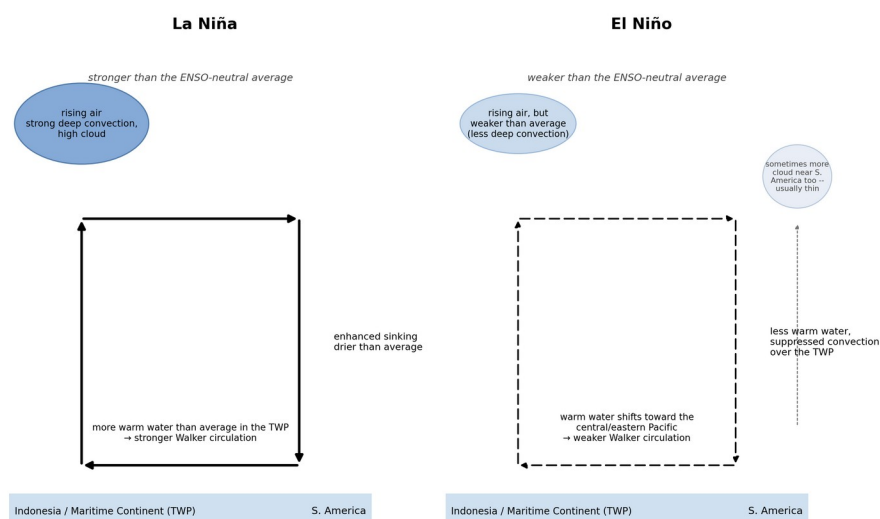


Figure 1. How ENSO changes the strength of the Walker circulation and tropical cloud. Schematic: course team.

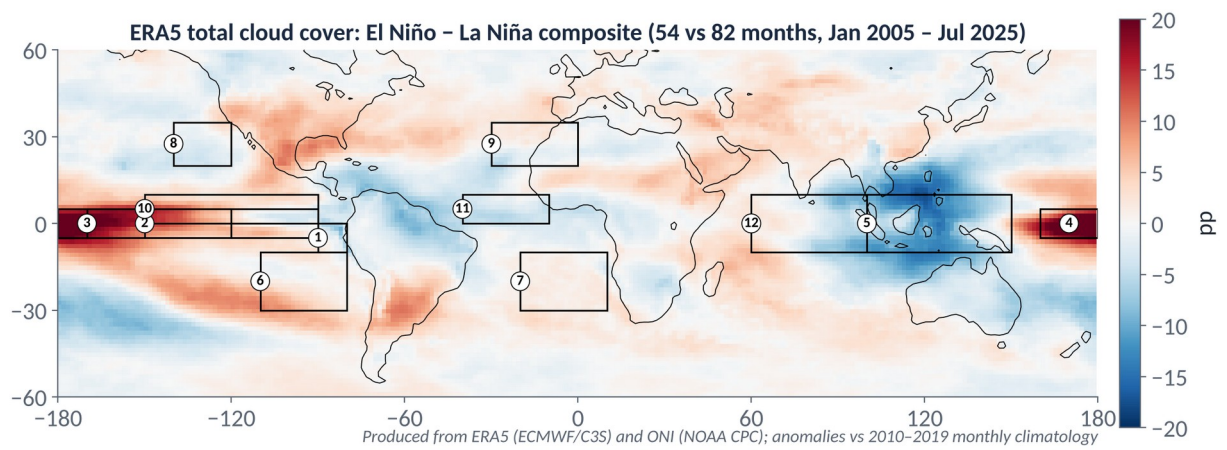


Figure 2. El Niño minus La Niña total cloud cover from the project's ERA5 data (54 El Niño vs 82 La Niña months, January 2005 - July 2025; anomalies relative to the 2010-2019 monthly climatology). Numbered boxes are the notebook's regions. Produced for this course from ERA5 (ECMWF/C3S) and ONI (NOAA CPC).

3. The data and the notebook

Launch `student_project.ipynb` on Binder (link on Canvas) and run it **from the top, in order**. Binder sessions are temporary: download your figures, results and notebook before closing.

Section	What it does
0 • 0b	Settings (variable, years, baseline, threshold) and loading the data (run once)
1	Global seasonal cycle
2	Global maps: El Niño – La Niña composite and correlation with the ONI
3 (3b)	Global-mean anomalies by ENSO phase (3b, optional: is one year different from another?)
4	Regional anomalies for 12 numbered regions
5	ONI vs regional anomaly: scatter plots with r , R^2 and n
6	Optional: correlation table with n , n_{eff} , 95% range and adjusted p ; lag and seasonal correlations

Settings you can change (section 0)

Setting	Default	Meaning
VARIABLE	"tcc"	Variable for maps and regional analysis: tcc, lcc, mcc, hcc, tcwv, tclw, tciw, skt, ...
ANALYSIS_START/END	2005, 2025	Years analysed
CLIM_START/END	2010, 2019	Baseline period for the monthly climatology
ENSO_THRESH	0.5	$ \text{ONI} \geq \text{this } (^\circ\text{C})$ classes a month as El Niño or La Niña

The 12 regions (latitude °S/°N; longitude °W negative, °E positive)

#	Region	Bounds (latS, latN, lonW, lonE)
1	Niño 1+2	-10, 0, -90, -80
2	Niño 3	-5, 5, -150, -90
3	Niño 3.4	-5, 5, -170, -120
4	Niño 4 (central Pacific)	-5, 5, 160, -150 (crosses the dateline)
5	Maritime Continent	-10, 10, 100, 150
6	SE Pacific stratocumulus	-30, -10, -110, -80
7	SE Atlantic stratocumulus	-30, -10, -20, 10
8	NE Pacific stratocumulus	20, 35, -140, -120
9	NE Atlantic stratocumulus	20, 35, -30, 0
10	ITCZ Pacific	0, 10, -150, -90

#	Region	Bounds (latS, latN, lonW, lonE)
11	ITCZ Atlantic	0, 10, -40, -10
12	Equatorial Indian Ocean	-10, 10, 60, 100

You can add your own region in the REGIONS setting. Give the bounds in your Methods.

4. The methods in brief

Method	What it is	Watch out for
Area weighting	Grid boxes shrink towards the poles (area $\propto \cos$ latitude), so means are weighted: $\bar{X} = \sum \cos(\varphi_i)X_i / \sum \cos(\varphi_i)$	Matters hugely for global means, little for narrow tropical boxes; say you did it
Climatology and anomaly	Climatology = mean for each calendar month over the baseline (2010–2019). Anomaly = value – climatology for that month	Positive means ‘cloudier than usual for that place and month’, not ‘cloudy’; state the baseline
Thresholds	Months with ONI $\geq +0.5$ °C are El Niño, ≤ -0.5 °C La Niña (54 and 82 months, 2005–2025)	NOAA's official episodes also need ≥ 5 overlapping seasons; one event (2015–16) supplies 14 El Niño months
Composite	Mean anomaly in El Niño months minus mean anomaly in La Niña months (units: pp)	A difference between two states, not a correlation; compare each phase with neutral to spot asymmetry
Correlation r	How consistently cloud follows the ONI along a straight line (–1 to +1, no units). $R^2 = r^2$ is the fraction of variance explained	Not the size of the effect; misses one-sided or curved responses; always plot the scatter
Regression slope	Change in cloud per °C of ONI (pp per °C)	Pulled by extreme months; straight line only
Significance	p = chance of a correlation at least this strong if there were no real relationship: $t = r\sqrt{(n-2)}/\sqrt{(1-r^2)}$, $n-2$ degrees of freedom	Needs n ; monthly ENSO data are autocorrelated, so use $n_{\text{eff}} = n(1 - r_{1x}r_{1y})/(1 + r_{1x}r_{1y})$ (notebook §6)
Lag	Correlation with the ONI shifted in time (notebook §6: positive lag = ONI leads)	Peaks are broad; a lag is consistent with cause but doesn't prove it

TWO TRAPS WORTH KNOWING

Many tests, some false positives. On a map with thousands of grid points, about 5% pass a 5% test by chance, and more when data are autocorrelated. Choose your region for a **physical** reason, then test it.

Trends. Global cloud in this ERA5 record falls by about 0.4 pp per decade. Years far apart can differ because of the trend, not ENSO; notebook §3b can remove it.

The *methods_demos.ipynb* notebook (optional, on Canvas) lets you try each of these ideas in a few seconds.

5. Writing the report

Your report is **one argument**, organised around your question, not a diary of the notebook. Every section and figure should help answer the question.

Section	≈ Words	Its job
Title		Informative and accurate: it should not promise more than you show
Introduction	350	Context the reader accepts → the problem ('however...') → "The purpose of this report is to..." (+ hypothesis if useful)
Methods	300	Data, period, region bounds, ONI and threshold, baseline, weighting, statistics actually used, with n. Science, not code; enough for a classmate to repeat it
Results	650	One paragraph per finding: the finding first, then quantified evidence citing the figure, e.g. "(Fig. 2)"
Discussion	450	Physical mechanism; comparison with published work; exceptions; limitations that matter; back to the question. Flag speculation
Conclusion	150	Answer the question in the first sentence; 2–3 findings with key numbers; nothing new
References		One consistent author–year style; include ERA5 and the ONI

Word counts are a guide. Your tutor will confirm whether captions and references count towards the 2,000 words.

The paragraph pattern

Claim → Evidence → Interpretation → Qualification. State the finding, give the numbers and the figure, explain it physically, then say where it doesn't hold.

SAME RESULT, TWO PARAGRAPHS (ILLUSTRATIVE NUMBERS)

× "Figure 3 shows that cloud cover changes during El Niño. There is a difference between the two phases. This could be due to ENSO affecting the atmosphere."

✓ "High-cloud cover over the central Pacific (Niño 4 region) averages about 14 percentage points above normal in El Niño months and 12 below in La Niña months (Fig. 3), and is strongly correlated with the ONI ($r = 0.8$, $n = 247$ months). This is consistent with the weakening of the Walker circulation during El Niño, when deep convection shifts eastward from the Maritime Continent. The response is not uniform, however: [n] of the [N] El Niño events show a much weaker signal, suggesting that event strength matters as well as ONI alone."

Citing data and literature

- ERA5: Hersbach, H. et al. (2020) The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.*, 146, 1999–2049.
- ONI: NOAA Climate Prediction Center (give the page and the date you accessed it).
- Find literature with Google Scholar and Web of Science; manage it with Zotero or EndNote (see the workbook's literature toolkit). Synthesise sources around ideas; don't list them one by one.

6. Figures, numbers and captions

Figures

- Include a figure because it is **evidence** for your answer, not because the notebook produced it. A typical set: one context figure, one or two that test the question, one that qualifies or compares.
- Label axes with variable and units. For anomalies, differences and correlations, use a **diverging colour scale centred on zero** with symmetric limits; panels you compare share the same limits.
- Cite every figure in the text, in numerical order, describing the **science**, not the plot: “Cloud cover over the Maritime Continent is lower in El Niño months (Fig. 2)”, not “Figure 2 shows...”.

Captions: the five-point check

A caption tells the reader how to read the figure without the main text. Begin with a phrase that captures its essence, then give:

- **What** is plotted (raw field, anomaly, difference, composite, correlation)
- **Units**
- **Period** and **region**
- Any **threshold or definition** (e.g. how El Niño months were chosen)
- The **data source(s)**, with the attribution line

EXAMPLE

× “Figure 2. Correlation map.”

✓ “Figure 2. Correlation between monthly high-cloud-cover anomaly and the ONI at each grid point, January 2005 – July 2025 (n = 247 months). Boxes mark the regions analysed in Figs 3–4. Figure produced by the author using ERA5 (ECMWF / Copernicus Climate Change Service) and ONI (NOAA Climate Prediction Center) data.”

Reporting numbers

- Give the number, not just the direction: “8 percentage points lower”, not “lower”.
- Match precision to the data (two decimal places for r is plenty), and state units, the baseline and the sample size (n, and n_eff if you test significance).
- Only use ‘significant’ if you ran a test, and say which. Keep correlation and causation apart: ‘is associated with’ unless you argue the mechanism.
- Sanity-check numbers copied from the notebook (an r outside –1 to +1 means something has gone wrong).

7. How your report is marked

The report is marked on a UK percentage scale. A report that does everything asked of it, correctly, sits at roughly **65%**, with real room above that for independent thinking.

Band	What it looks like
40–49%: Pass / Third	A question is present but vague or broad, or not clearly answered. Some figures, but they may be poorly chosen, mislabelled or largely descriptive. The report tends to follow the notebook's narrative rather than build an argument. Basic concepts are used, sometimes with misunderstandings. Sources are at least partly acknowledged.
50–59%: 2:2	One reasonably specific question is answered, though the reasoning from evidence to conclusion may be thin. Figures are mostly relevant and well presented, but one or two may not serve the question. Some quantification, perhaps imprecise. The standard structure is followed, though sections may overlap. Attribution is present and mostly correct.
60–69%: 2:1 (solid, competent; roughly 65%)	A genuinely focused question, clearly stated and directly answered in the Conclusion. Three to five well-designed figures, each earning its place as evidence. Claims consistently backed by quantified evidence (sign, magnitude, r/R^2 , n) and physically interpreted. Some awareness of limitations. Clean structure: a funnel Introduction, Methods that explain what was done and why (not a Python how-to), Results and Discussion kept separate, and a Conclusion with no new analysis.
70–79%: First	A well-chosen, specific, physically motivated question. Excellent figures, with physical insight discussed rather than just described. Claim → Evidence → Interpretation → Qualification used consistently, with thoughtful qualifications. Independent scientific thinking in the Discussion: comparing regions or methods, relating results to the Walker circulation, SST, subsidence and convection, noting what was surprising or expected. Concise writing; complete, precise attribution.
80%+: Exceptional	All of the above, plus something beyond what was directly taught: an original angle, an insightful comparison, or a careful and correct treatment of a subtlety (e.g. effective sample size, a sensitivity test). Writing and figures of near-publishable short-communication standard. Rare.

WHAT PULLS A MARK DOWN, WHATEVER THE BAND

No single clear question • including every notebook output rather than choosing evidence • treating correlation as proof of causation • unattributed data or images • code in the report • new analysis in the Conclusion.

8. Common pitfalls

- A question so broad it becomes 'ENSO and all clouds everywhere'.
- Writing Results in the order you ran the notebook, rather than around the question.
- Describing colours on a map without giving the sign, size and location of the signal.
- A correlation with no n, period or test; 'significant' without a test.
- Percent when you mean percentage points.
- Physical interpretation in the Methods (it belongs in the Discussion); Python in the Methods (describe the science).
- A region box that straddles opposite signals, so the average shows nothing.
- Forgetting that one big event (2015–16) can dominate a composite.
- Forgetting to cite ERA5 and the ONI.

Key dates and resources

When	What
Thursdays 1 Oct – 3 Dec	Tutorials 1–8 (no tutorials on 22 Oct or 5 Nov)
Thu 3 Dec 2026	Complete first draft brought to Tutorial 8
Fri 5 Feb 2027	Draft submitted (formative; no DASS extensions; your tutor confirms how)
w/c 8 or 15 Feb 2027	Peer feedback on drafts
w/c 1 Mar 2027	Group presentation
Fri 12 Mar 2027, 12:00	Final report submitted (DASS extensions apply)
Wed 7 Apr 2027	Marks and feedback released

On Canvas: this guide, the Writing Workbook, the tutorial slides (W1–W8), Binder links for both notebooks, and the submission link.

Data: ERA5, ECMWF / Copernicus Climate Change Service; ONI, NOAA Climate Prediction Center. *Schematics:* course team.